Tech+Ball BioGrip™ Smart Ball Deliverable Sheet

Manufacturing Specifications, Fact-Checked Design Audit, Cost Model & R&D Roadmap

1. EXECUTIVE SUMMARY & SMARTBALL DESIGN AUDIT

Design Evolution & Verification Audit

Fact Check Audit Summary: The Tech+Ball smartball design evolved from an earlier *3D Sonic Fingerprint Sensor* design (rigid shell with optical/ultrasonic TFT glass scanners) to the patented **BioGrip™ Sensor Fabric** (conductive fibers woven into a Kevlar base fabric with clear silicone composite encapsulation). This touch-sensory smart fabric replaces fragile glass/optical components, capturing athlete hand bio-ridges, pressure distribution, palm prints, and grip dynamics directly during high-impact field play.

The **Tech+Ball BioGrip™ Smart Ball** serves as the core telemetry transceiver for the NIBLS (National Intelligent Basketball/Ball League Sport) decentralized sports fintech ecosystem. The ball records real-time physical metrics (passes, catches, scores, velocity, rotational spin, throwing power, and hand-strip steals) and streams 1000Hz telemetry to the cloud for real-time **ALT Team Token** market valuation and algorithmic sports trading.

Table 1.1: Smartball Design Evolution & Architecture Fact Check

Design Parameter	Legacy Design (3D Sonic Version)	BioGrip™ Sensor Fabric Design (Current)
Biometric Layer	Ultra-thin sonic film transistor (TFT) 3D scanner	BioGrip™ conductive fibers woven into Kevlar base fabric
Touch Recognition	Glass/optical ultrasonic surface scanner	Detects bio ridges, valleys, palm prints & grip pressure distribution
Outer Shell	ABS / Ninja Flex / Rigid Silicone	Clear protective silicone composite coating over conductive Kevlar
Durability & Field Use	Susceptible to glass fracture upon hard impacts	Encapsulated flexible fabric; withstands turf, grass, & concrete impact
Chassis Dimensions	7.48 in x 7.48 in x 7.48 in (31.45 in^3)	7.48 in x 7.48 in x 7.48 in (31.45 in^3) regulation spherical shell
Microchip Suspension	Standard internal mounting bracket	NIBLS Core-v4 Microchip suspended at mathematical bladder center

2. LAYERED HARDWARE ARCHITECTURE & ONBOARD COMPUTING

The smart ball integrates five concentric physical layers paired with an onboard micro-computing array to collect and transmit high-frequency telemetry at 1000Hz without disturbing ball balance or aerodynamic flight.

Layer / Subsystem	Material / Component	Functional Duty & Technical Spec
Outer Protective Layer	Clear Silicone Composite Coating	Shields flexible circuitry & conductive fibers from moisture & turf impact
Biometric Base Layer	BioGrip™ Sensor Fabric	Conductive fibers in Kevlar fabric for palm/grip pattern recognition
Core Ball Shell	Ninja Flex / Kevlar / ABS / Rubber	Standard 31.45 in^3 spherical bladder chassis inflated to regulation PSI
Inner Middle Layer	Flexible Printed Circuit Board (PCB)	Data processing board mounted with silver connector plates & insert screws
Compute Engine	Raspberry Pi (Pi B+ / Pi 3)	Manages Wi-Fi, Bluetooth, onboard ball computations & wireless streaming
Speed & Motion	ADXL345 Accelerometer + Gyroscope	Tracks linear velocity, acceleration & rotational spin (up to 3,600°/sec)
Distance & Power	GPS Sensor + Ultrasonic Ranger	Measures midair flight distance (Arduino Uno) & kinetic throwing force
Goal & Force Sensors	Motion Tracker + Force Sensors	Detects goal line crossings & hand grip pressure distribution
Power Source	Internal Rechargeable Ion Battery	Powers internal microchip, computing unit, sensors & diagnostic LCD

3. MANUFACTURING COST BREAKDOWN & PRICING STRATEGY

NIBLS, Inc. maintains a strict bill of materials (BOM) cost model engineered for high gross margins while ensuring accessibility for local franchisee clubs and professional leagues.

Component / Cost Element	Category	Unit Cost ($ USD)	% of Total Cost	Notes & Sourcing Specifications
BioGrip™ Sensor Fabric	Materials	$20.00	36.4%	Conductive fibers woven into Kevlar base layer
Flexible Circuit Board (PCB)	Electronics	$10.00	18.2%	Custom inner middle layer PCB with silver plates
Additional Sensor Suite	Sensors	$7.00	12.7%	ADXL345, Gyroscope, GPS, Ultrasonic & Force sensors
Core Ball Material	Materials	$5.00	9.1%	Kevlar / ABS / Ninja Flex bladder shell
Rechargeable Ion Battery	Power	$5.00	9.1%	Internal ion battery powering 32-hr development runs
Protective Coating	Materials	$3.00	5.5%	Clear silicone composite protective coating
Direct Assembly Labor	Labor	$10.00	18.2%	Precision mounting, bladder balancing & soldering
Manufacturing Overhead	Overhead	$5.00	9.1%	Quality control, calibration & factory overhead
TOTAL MANUFACTURING COST	**BOM Total**	**$55.00**	**100.0%**	**Target Unit Cost across Phase 4 Production Run**

Commercial Pricing & Monetization Model

Commercial Pricing & Business Model: Retail Price: $150.00 / unit ($95.00 gross margin per ball, ~63.3% margin). | **Hardware Bundle Price:** $350.00+ (Includes Smart Ball, Motion Sensor Goal Device & ECG Heart Rate E-Wallet). | **Wholesale Price:** $100.00 / unit (Bulk team/league franchisee package orders). | **Data & Wi-Fi Service:** $50.00 / week maintenance fee for live telemetry streaming & NBL token analytics.

4. SOFTWARE ARCHITECTURE & DATA PIPELINE INTEGRATION

Telemetry captured by the BioGrip™ Smart Ball passes through a multi-tier cloud and blockchain pipeline to guarantee real-time performance valuation without database latency:

• **Embedded Firmware:** Written in C, C++, Python, or C# using static memory block allocation to eliminate dynamic allocation overhead, driving the interior LCD and streaming 1000Hz telemetry.

• **Cloud Ingestion Pipeline:** Feeds stream via *Azure IoT Hub / AWS IoT* to *Azure Streaming Analytics / AWS Lambda* microservices, storing data in *Azure Cosmos DB / Amazon S3* via *Azure Private Link*.

• **Blockchain Infrastructure:** Interfaces with *Hyperledger Fabric* via RESTful APIs and smart contracts for tokenized ticketing, athlete skill verification, and proof-of-data logging.

• **Sharding Algorithmic Engine:** Horizontally partitioned distributed database executing rule-based query scanning and index optimization to strictly avoid sequential table scans. Physical field metrics (distance, passes, catches, scores, speed, power, steals) are automatically translated into real-time **ALT Team Token** value adjustments.

5. MANUFACTURING TIMELINE (05/2024 – 01/2026) & R&D ROADMAP

Phase / Stage	Timeline	Core Deliverables & Milestones
Phase 1: Conceptual Design	05/2024 – 09/2024	3D CAD modeling, 3D printed smart ball mockups for component mounting, off-the-shelf component sourcing.
Phase 2: Exterior & Interior Eng.	10/2024 – 03/2025	Carbon fiber / Ninja Flex / clear silicone gel ball engineering, interior PCB mounting, firmware development for LCD readouts.
Phase 3: Sensor & Circuitry	04/2025 – 08/2025	BioGrip™ fabric integration, biometric fingerprint sensor assemblies, circuit board functionality & communication loop testing.
Phase 4: Full App & Custom PCB	09/2025 – 01/2026	Full application software (two-way computer communication), custom PCB transition, functional video demos, engineering minutes & charcoal carbon fiber packaging.
R&D Testing Cycle (7-Stage)	Ongoing Week 1-2	3D printing → Hardware Code -> Hardware Testing -> Blockchain Code -> Blockchain Testing -> System Integration -> Brand Licensing (Adidas/Nike/Molten).
Field Testing & Exit	Data Phase	Deploy smart balls across local seasons & playoff challenges to build athletic datasets; execute M&A corporate exit.

Authorization & Specification Verification
Document Sign-Off & Engineering Authorization:
• **Founder & Hardware Inventor:** Dashawn R. Bledsoe (NIBLS, Inc.)
• **Lead Mechanical Engineer:** JC (University of Buffalo Graduate / NIBLS, Inc.)
• **Status:** Fact-Checked & Approved for Phase 4 Manufacturing Deliverables (Target Completion: January 2026).